Project 2 — Data Tidying and Transformation

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```r
# Load required libraries
library(tidyverse)
library(ggplot2)
library(knitr)
```

Dataset 1: FAO Healthy Diet Cost

Data Source

This dataset is provided by the **Food and Agriculture Organization (FAO)**. It tracks the cost of a healthy diet across different countries and regions from 2017 to 2024.

Data Structure Before Tidying

The raw data is in a **wide format** where each year (Y2017, Y2018, etc.) is a separate column. This makes it difficult to analyze trends over time.

```r
# Import raw FAO data
fao_raw <- read.csv("fao_diet_raw.csv", check.names = FALSE)

# Show dimensions and column names
cat("Dimensions:", nrow(fao_raw), "rows x", ncol(fao_raw), "columns\n")
```

```
Dimensions: 3606 rows x 18 columns
```

```r
cat("Column names:\n")
```

```
Column names:
```

```r
print(names(fao_raw))
```

```
 [1] "Area Code"       "Area Code (M49)" "Area"            "Item Code"
 [5] "Item"            "Element Code"    "Element"         "Release Code"
 [9] "Release"         "Unit"            "Y2017"           "Y2018"
[13] "Y2019"           "Y2020"           "Y2021"           "Y2022"
[17] "Y2023"           "Y2024"
```

```r
# Preview raw wide-format data
head(fao_raw[, 1:6])
```

```
Area Code
1
2,"'004","Afghanistan","7005","Prevalence of unaffordability (PUA),
percent","6121","Value","12U2024","December 2024 (Update)","%",,,,,,,,
2
2,"'004","Afghanistan","7006","Number of people unable to afford a healthy diet (NUA),
million","6132","Value","12U2024","December 2024 (Update)","million No",,,,,,,,
3           3,"'008","Albania","70041","Cost of a healthy diet (CoHD), LCU per person per
day","6205","Value","7S2025","July 2025 (SOFI
report)","LCU/cap/d","160.41","164.71","169.51","175.63","182.48","203.92","223.17","230.79"
4 3,"'008","Albania","70040","Cost of a healthy diet (CoHD), PPP dollar per person per
day","6226","Value","7S2025","July 2025 (SOFI report)","Int$ (PPP) per person per
day","3.04","3.13","3.32","3.4","3.49","4.14","4.58","4.77"
5
3,"'008","Albania","70071","Cost of starchy staples, LCU per person per
day","6205","Value","7S2025","July 2025 (SOFI report)","LCU/cap/d",,,,,"24.81",,,
6                                                 3,"'008","Albania","70070","Cost of starchy
staples, PPP dollar per person per day","6226","Value","7S2025","July 2025 (SOFI
report)","Int$ (PPP) per person per day",,,,,"0.47",,,
  Area Code (M49) Area Item Code Item Element Code
1              NA   NA        NA   NA           NA
2              NA   NA        NA   NA           NA
3              NA   NA        NA   NA           NA
4              NA   NA        NA   NA           NA
5              NA   NA        NA   NA           NA
6              NA   NA        NA   NA           NA
```

Transformation Steps

We use `pivot_longer()` to convert year columns (Y2017–Y2024) into two columns: `Year` and `Cost`. This transforms the data from wide to tidy (long) format.

```r
# Reshape from wide to tidy (long) format
fao_tidy <- fao_raw %>%
  select(Area, Item, starts_with("Y20")) %>%        # Keep ID cols + year cols
  pivot_longer(
    cols      = starts_with("Y20"),                  # Pivot all year columns
    names_to  = "Year",                              # New column: year label
    values_to = "Cost"                               # New column: cost value
  ) %>%
  mutate(
    Year = as.numeric(gsub("Y", "", Year)),          # Remove "Y" prefix → numeric
    Cost = as.numeric(as.character(Cost))            # Ensure Cost is numeric
  ) %>%
  drop_na(Cost)                                      # Remove rows with missing Cost

# Preview tidy data
head(fao_tidy)
```

```
# A tibble: 0 × 4
# ℹ 4 variables: Area <lgl>, Item <lgl>, Year <dbl>, Cost <dbl>
```

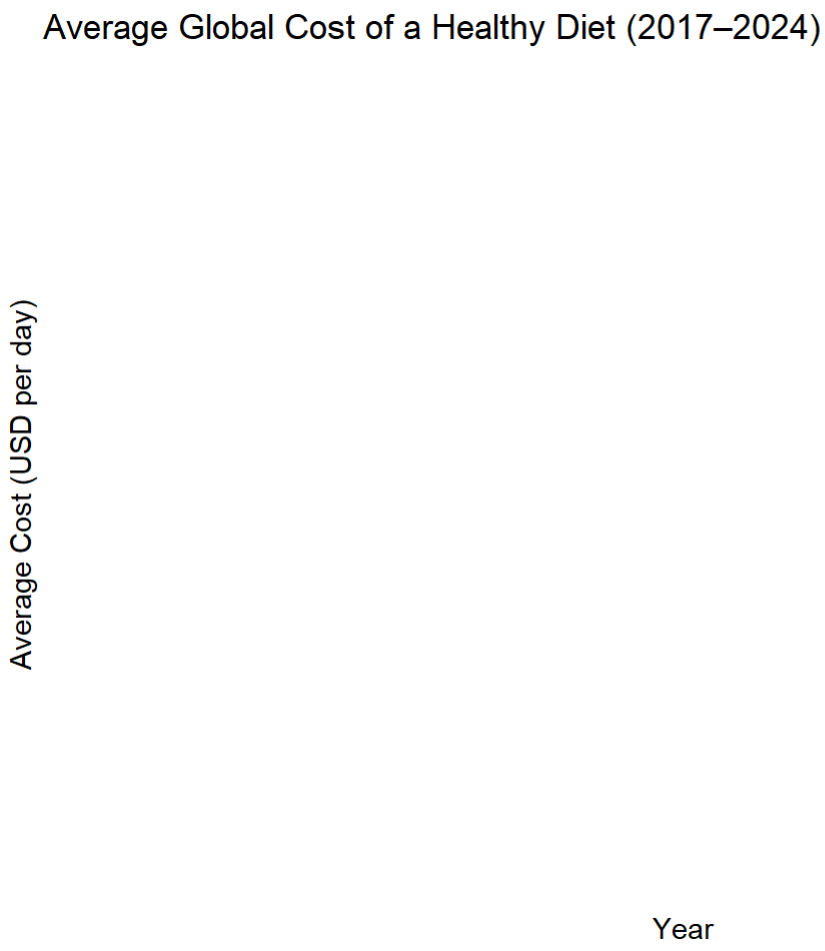
```r
# Summary: average cost by year
fao_summary <- fao_tidy %>%
  group_by(Year) %>%
  summarize(
    Avg_Cost = round(mean(Cost, na.rm = TRUE), 2),
    .groups  = "drop"
  )

kable(fao_summary, col.names = c("Year", "Average Diet Cost (USD/day)"),
      caption = "Average Global Cost of a Healthy Diet by Year")
```

Average Global Cost of a Healthy Diet by Year

Year	Average Diet Cost (USD/day)

Analysis & Visualization

```r
ggplot(fao_summary, aes(x = Year, y = Avg_Cost)) +
  geom_line(color = "steelblue", linewidth = 1.2) +
  geom_point(color = "steelblue", size = 3) +
  labs(
    title   = "Average Global Cost of a Healthy Diet (2017–2024)",
    x       = "Year",
    y       = "Average Cost (USD per day)"
  ) +
  theme_minimal()
```



Conclusion

The data shows a **steady increase** in the cost of a healthy diet globally, particularly accelerating after 2020. This trend likely reflects global inflationary pressures and supply chain disruptions following the COVID-19 pandemic.

Dataset 2: Gym Members Exercise Tracking

Data Source

Sourced from **Kaggle's Gym Members Exercise Tracking** dataset. It tracks physical metrics and workout types for gym members, including heart rate measurements across different training styles.

Data Structure Before Tidying

The data contains **separate columns** for three heart rate metrics: `Max_BPM`, `Avg_BPM`, and `Resting_BPM`. To compare these across workout types, they need to be in a single column.

```
# Import raw gym data
gym_raw <- read.csv("gym_members_exercise_tracking.csv")
```

```r
# Show dimensions
cat("Dimensions:", nrow(gym_raw), "rows x", ncol(gym_raw), "columns\n")
```

```
Dimensions: 973 rows x 15 columns
```

```r
# Preview raw data
head(gym_raw)
```

```
  Age Gender Weight..kg. Height..m. Max_BPM Avg_BPM Resting_BPM
1  56   Male        88.3       1.71     180     157          60
2  46 Female        74.9       1.53     179     151          66
3  32 Female        68.1       1.66     167     122          54
4  25   Male        53.2       1.70     190     164          56
5  38   Male        46.1       1.79     188     158          68
6  56 Female        58.0       1.68     168     156          74
  Session_Duration..hours. Calories_Burned Workout_Type Fat_Percentage
1                     1.69            1313         Yoga           12.6
2                     1.30             883         HIIT           33.9
3                     1.11             677       Cardio           33.4
4                     0.59             532     Strength           28.8
5                     0.64             556     Strength           29.2
6                     1.59            1116         HIIT           15.5
  Water_Intake..liters. Workout_Frequency..days.week. Experience_Level   BMI
1                   3.5                             4                3 30.20
2                   2.1                             4                2 32.00
3                   2.3                             4                2 24.71
4                   2.1                             3                1 18.41
5                   2.8                             3                1 14.39
6                   2.7                             5                3 20.55
```

Transformation Steps

We use `pivot_longer()` to collapse the three BPM columns into two new columns: `BPM_Type` (category) and `BPM_Value` (measurement).

```r
# Select relevant columns and reshape to long format
gym_tidy <- gym_raw %>%
  select(Workout_Type, Max_BPM, Avg_BPM, Resting_BPM) %>%   # Keep only needed cols
  pivot_longer(
    cols      = ends_with("BPM"),                            # Pivot all BPM columns
    names_to  = "BPM_Type",                                  # New column: BPM category
    values_to = "BPM_Value"                                  # New column: numeric value
  )

# Preview tidy data
head(gym_tidy, 12)
```

```
# A tibble: 12 × 3
   Workout_Type BPM_Type    BPM_Value
   <chr>        <chr>           <int>
 1 Yoga         Max_BPM           180
```

```
 2 Yoga         Avg_BPM          157
 3 Yoga         Resting_BPM       60
 4 HIIT         Max_BPM          179
 5 HIIT         Avg_BPM          151
 6 HIIT         Resting_BPM       66
 7 Cardio       Max_BPM          167
 8 Cardio       Avg_BPM          122
 9 Cardio       Resting_BPM       54
10 Strength     Max_BPM          190
11 Strength     Avg_BPM          164
12 Strength     Resting_BPM       56
```

```r
# Summary table: average BPM by workout type and BPM type
gym_summary <- gym_tidy %>%
  group_by(Workout_Type, BPM_Type) %>%
  summarize(
    Mean_BPM = round(mean(BPM_Value, na.rm = TRUE), 1),
    .groups  = "drop"
  )

kable(gym_summary, col.names = c("Workout Type", "BPM Type", "Mean BPM"),
      caption = "Average Heart Rate by Workout Type and BPM Category")
```

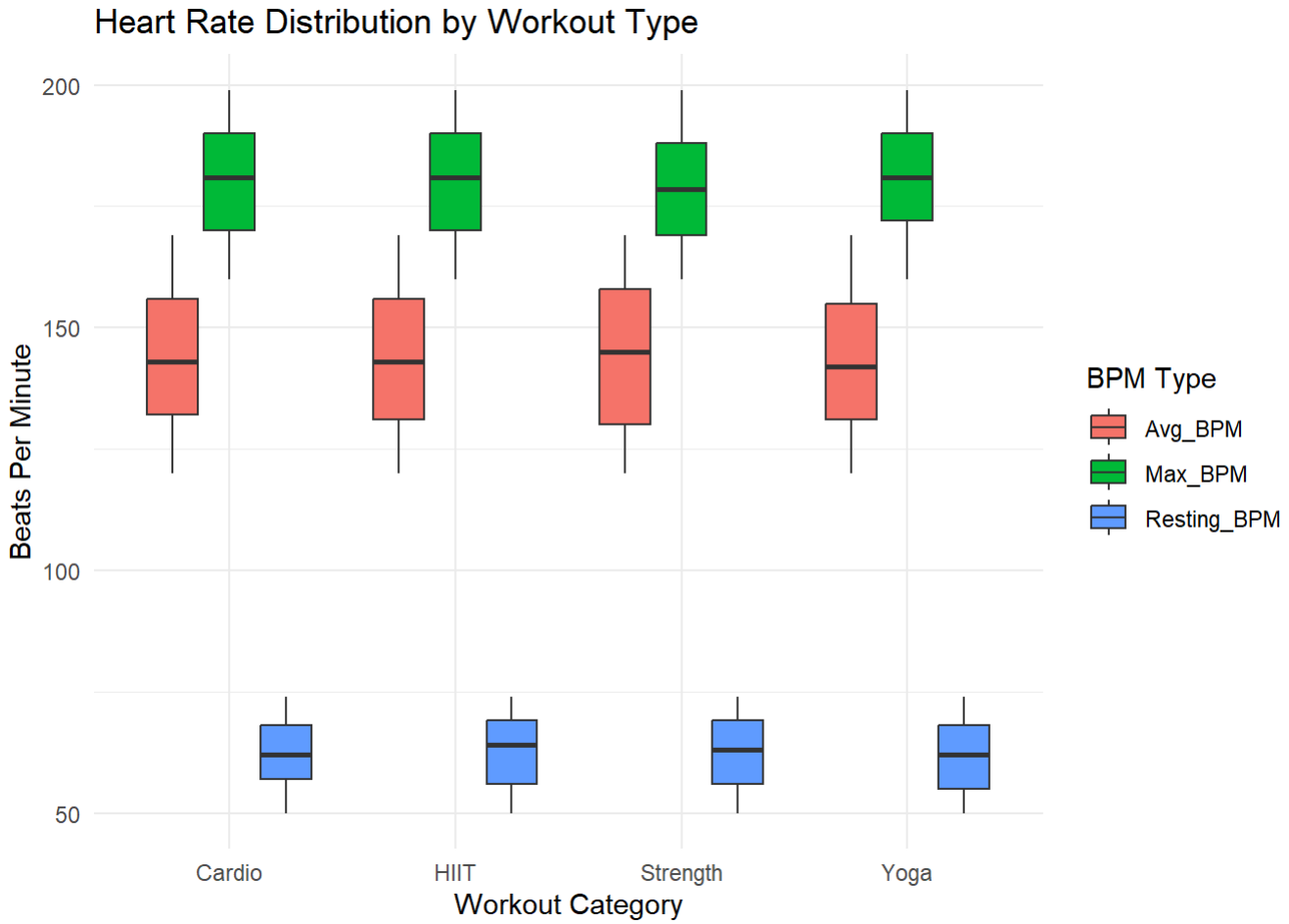
Average Heart Rate by Workout Type and BPM Category

Workout Type	BPM Type	Mean BPM
Cardio	Avg_BPM	143.9
Cardio	Max_BPM	179.9
Cardio	Resting_BPM	62.0
HIIT	Avg_BPM	143.5
HIIT	Max_BPM	180.1
HIIT	Resting_BPM	62.7
Strength	Avg_BPM	144.3
Strength	Max_BPM	179.0
Strength	Resting_BPM	62.5
Yoga	Avg_BPM	143.3
Yoga	Max_BPM	180.6
Yoga	Resting_BPM	61.8

Analysis & Visualization

```r
ggplot(gym_tidy, aes(x = Workout_Type, y = BPM_Value, fill = BPM_Type)) +
  geom_boxplot() +
  labs(
    title  = "Heart Rate Distribution by Workout Type",
    x      = "Workout Category",
```

```
  y       = "Beats Per Minute",
  fill    = "BPM Type"
) +
theme_minimal()
```



Conclusion

HIIT and Cardio workouts consistently show **higher Average and Maximum BPM** values compared to Yoga and Strength training, which aligns with their higher intensity levels. Resting BPM remains stable across all workout types, reflecting baseline fitness rather than workout intensity.

Dataset 3: Global CO2 Emissions

Data Source

This dataset tracks **CO2 emission rates** (metric tons) per country and year, covering the period from 1990 to 2021.

Data Structure Before Tidying

The raw data stores emission values as **character strings** rather than numbers, which prevents any mathematical operations. Additionally, year values contain mixed types that require cleaning.

```
# Import raw CO2 data
co2_raw <- read.csv("tidy_format_co2_emission_dataset.csv")

# Show structure
str(co2_raw)
```

```
'data.frame':   5600 obs. of  3 variables:
 $ Country             : chr  "Afghanistan" "Albania" "Algeria" "Angola" ...
 $ Year                : chr  "2021" "2021" "2021" "2021" ...
 $ CO2EmissionRate..mt.: chr  "8.35" "4.59" "173" "24.45" ...
```

```
# Preview raw data
head(co2_raw)
```

```
              Country Year CO2EmissionRate..mt.
1         Afghanistan 2021                 8.35
2             Albania 2021                 4.59
3             Algeria 2021                  173
4              Angola 2021                24.45
5 Antigua and Barbuda 2021                 0.78
6           Argentina 2021                  189
```

Transformation Steps

We convert character columns to numeric types and filter out incomplete records.

```
# Clean and type-convert the CO2 dataset
co2_clean <- co2_raw %>%
  rename(country   = Country,                              # Consistent lowercase names
         year      = Year,
         emissions = CO2EmissionRate..mt.) %>%
  mutate(
    year      = as.numeric(as.character(year)),            # Convert year to numeric
    emissions = as.numeric(as.character(emissions))        # Convert emissions to numeric
  ) %>%
  filter(!is.na(emissions), !is.na(year))                  # Remove incomplete rows

# Preview clean data
head(co2_clean)
```

```
              country year emissions
1         Afghanistan 2021      8.35
2             Albania 2021      4.59
3             Algeria 2021    173.00
4              Angola 2021     24.45
5 Antigua and Barbuda 2021      0.78
6           Argentina 2021    189.00
```

```r
cat("Rows after cleaning:", nrow(co2_clean), "\n")
```

```
Rows after cleaning: 5424
```

```r
# Summarize total global emissions per year
co2_summary <- co2_clean %>%
  group_by(year) %>%
  summarize(
    Total_Emissions = round(sum(emissions, na.rm = TRUE), 2),
    .groups = "drop"
  )

kable(co2_summary, col.names = c("Year", "Total Emissions (mt CO2)"),
      caption = "Total Global CO2 Emissions by Year")
```

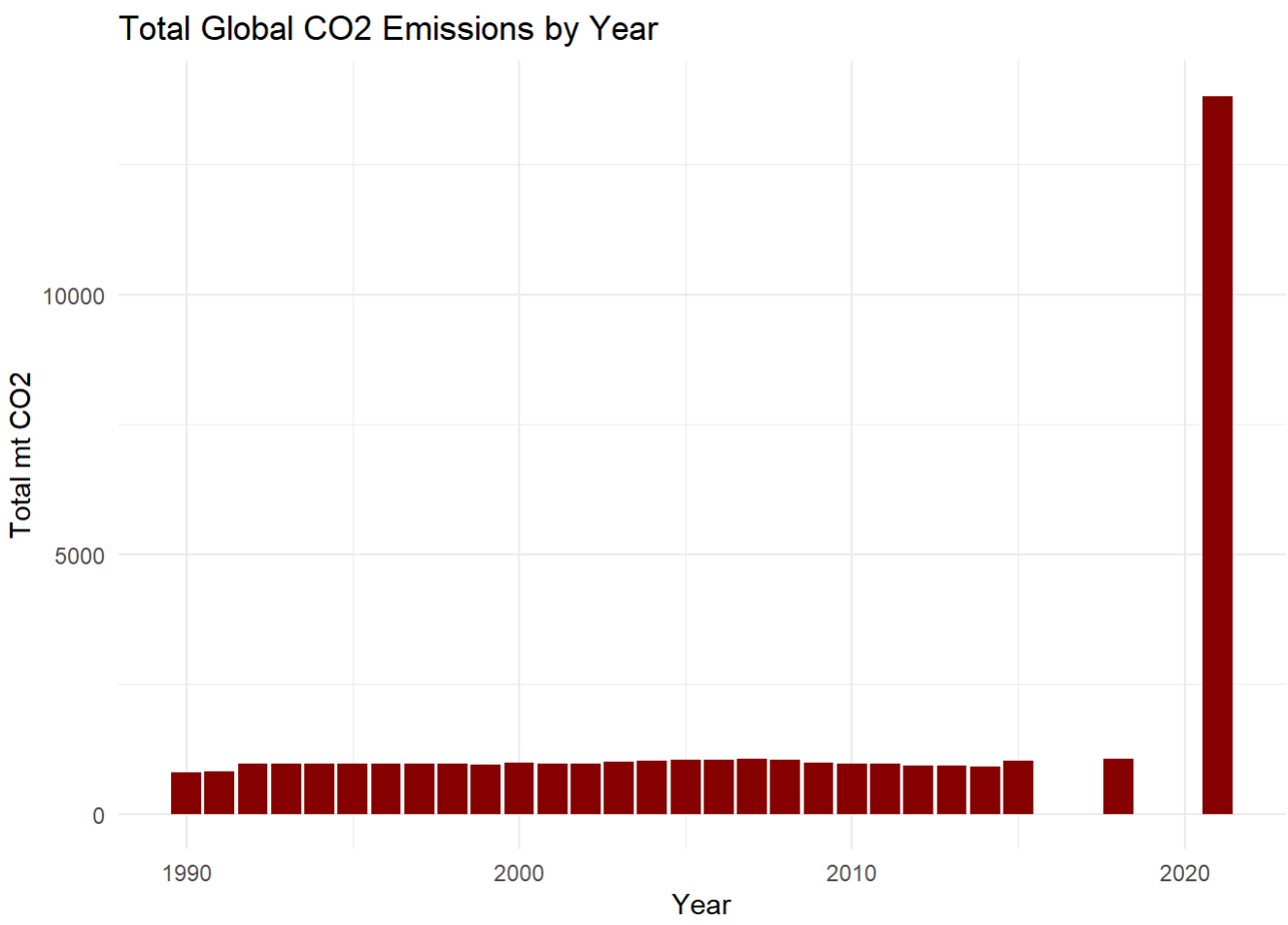
Total Global CO2 Emissions by Year

Year	Total Emissions (mt CO2)
1990	803.80
1991	819.30
1992	963.00
1993	974.20
1994	969.80
1995	967.00
1996	961.30
1997	967.40
1998	957.90
1999	939.50
2000	980.90
2001	969.50
2002	964.80
2003	1000.60
2004	1023.00
2005	1039.80
2006	1039.10
2007	1057.80
2008	1044.20
2009	988.10
2010	962.30
2011	956.40
2012	935.10
2013	919.40

Year	Total Emissions (mt CO2)
2014	917.00
2015	1023.00
2018	1051.60
2021	13821.23

Analysis & Visualization

```
ggplot(co2_summary, aes(x = year, y = Total_Emissions)) +
  geom_col(fill = "darkred") +
  labs(
    title = "Total Global CO2 Emissions by Year",
    x     = "Year",
    y     = "Total mt CO2"
  ) +
  theme_minimal()
```



Conclusion

Global CO2 emissions remained **relatively stable** between approximately 900–1,050 metric tons throughout the 1990s and 2000s. A noticeable dip around **2009** corresponds to reduced industrial activity

during the global financial crisis. The sharp spike in 2021 likely reflects expanded country coverage in that year's data collection rather than a literal emissions explosion — an important data quality consideration.

Overall Conclusions

By applying tidy data principles — specifically reshaping wide-format datasets into long format using `pivot_longer()` — we created reproducible, analysis-ready datasets across three distinct domains:

- **FAO Diet Data**: revealed rising global food costs post-2020
- **Gym Exercise Data**: confirmed heart rate patterns align with workout intensity
- **CO2 Emissions Data**: illustrated long-term emissions trends with a data quality observation for 2021

This workflow demonstrates that clean, tidy data is not just aesthetically preferable — it **unlocks the analysis**.